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Reinforcement Learning-based Multi-channel Random Access for Massive Machine-Type Communication in 5G Networks

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Abstract

This paper takes into account the development trend of random access methods in wireless communication networks, traffic features and random access methods for massive Machine-Type Communications (mMTC). Existing multi-channel random access methods based on artificial intelligence mechanisms use neural networks due to a large number of system states. Many mMTC applications require a simple implementation of the random access method and low power consumption of mMTC devices. The paper proposes a simple multi-channel random access method based on reinforcement learning, called RL-MCSA. The states of the system (radio channel) are the total number of collisions during access. Only the previous state of the radio channel is required to make a decision. The reward for the decision is the total number of successful accesses. The decision is made in order to maximize the total number of successful packet transmissions in frequency-time channels (resource blocks) in the uplink direction of the 5G access network. Simulation-based experiment results show that the proposed method can improve the rate of successful access in each simulation cycle and the rate of utilization of the radio channel by 1.5-2 times more than existing, well-known algorithms.

Keywords: 5G access networks, mMTC, random radio channel access, access probability, access collision, reinforcement learning

1 Introduction

In recent decades, there has been a trend towards active growth in the number of data transmission systems built on the basis of multiple access channels. Among the methods for controlling access of a large number of subscribers (devices, terminals) to a common channel (radio channel), random multiple access (RMA) methods occupy a special place. In RMA systems, a situation may arise in which two or more subscribers simultaneously use the same frequency-time resources of the channel. In this case, user signals interfere with each other, as a result of which none of the transmitted messages can be successfully received. Such an event is called a conflict. The first system to use random multiple access was the ALOHA system, created in the late sixties of the twentieth century for communication between computers at the University of Hawaii. The conflict resolution algorithm used in ALOHA was proposed and studied by N. Abramson [1]. At the end of the last decade of the last century, random multiple access received a new impetus in development due to its use in modern 5G wireless networks.

5G wireless services include three key services that address the needs of all aspects of life. Enhanced Mobile Broadband (eMBB) service aims to increase data transmission speeds to 10 gigabits per second (Gbps); massive machine-type communications (mMTC) aims to support the connection of billions of Internet of Things (IoT) devices; and ultra-reliable low-latency communications (URLLC) is a service class to support mission-critical applications [2, 3].

Massive machine-type communications is considered one of the fundamental services in the future digital world, enabling a wide range of applications in current 5G and future Beyond 5G (B5G) networks [4]. mMTC is designed to support communications between a huge number of devices, such as sensors and IoT (Internet of Things) devices. The density of mMTC devices (MTCD) will reach 106 per square kilometer in urban environments, which is significantly higher than that of human-type communication devices (HTC) [5]. According to Ericsson, the number of devices connected to communication networks will reach 26.9 billion by 2027, of which more than 50% will be MTCDs [6]. These devices often operate in resource-constrained environments and require efficient use of energy. Typical latency for mMTC is usually between 1 and 10 seconds, as most applications do not require instant data transfer. Examples of mMTC use cases include smart cities, environmental monitoring, and energy management systems.

In general, to support a large number of devices with infrequent activity, random access is widely considered since it avoids high signaling overhead. In particular, most mMTC schemes are based on slot ALOHA (S-ALOHA) and its modifications [7–9].

The basic idea of random access is that subscribers who have data to transmit send them to the channel with some probability p. The basic ALOHA algorithm, in which a subscriber, regardless of the channel situation, transmits data with probability p.

This idea is also used in p-CSMA (p - persistent) protocols [10]. Most algorithms are based on the same approach, the idea of which is as follows: a subscriber changes the transmission probability in some way depending on the events in the channel. The choice of the probability value p affects the probability of collision and the number of active devices. Too large a value of p leads to a larger number of devices and possible collisions, and too small a value leads to inefficient use of resources. In the case where the number of devices ready to transmit is unknown or changes over time, the optimal value of p cannot be set in advance.

In this paper, a method of multi-channel random access in the uplink of 5G access network is proposed based on reinforcement learning (RL), which is one of the mechanisms of artificial intelligence [11]. In the proposed method, the value of p is determined based on the control of the total number of collisions and the total number of successful transmissions in frequency-time channels is maximized. A comparative analysis of various p selection schemes shows that the proposed method increases the probability of successful access to the radio channel and reduces the average delay time of a data packet.

2 Related Works

The first version of single-channel ALOHA random access is also called *pure ALOHA* [12]. When using this method of channel access, user terminals begin transmitting data packets immediately after the information to be transmitted appears. If the transmission of two or more stations coincides in time, the base station cannot correctly receive the data. To give senders the opportunity to detect a collision, the base station broadcasts the received data packet after receiving it. By comparing the transmitted packet and the received one, the sender can understand whether its data was received correctly or with errors. If the data was transmitted incorrectly, the sender waits a random time interval and makes a second attempt to transmit.

Lawrence Roberts proposed another version of the single-channel ALOHA system, called slotted ALOHA [13]. The main difference between slotted ALOHA and pure ALOHA was the idea of dividing the time axis into discrete intervals of equal duration, called slots. When data packets intended for transmission appeared, the terminal delayed transmission until the beginning of the next slot. The duration of the slots was chosen so that during one slot the terminal had time to transmit its data packet and receive confirmation of successful transmission from the base station.

Single-channel ALOHA (binary exponential backoff, multiple access with eavesdropping, S-ALOHA, q-ALOHA) algorithms have simple implementation and excellent performance under light traffic conditions, but channel throughput decreases under high traffic.

Single-channel S-ALOHA is a basic scheme for random access control, and many random access methods to frequency-time channels (resource blocks) of modern wireless networks can be considered as its derivatives.

Deploying a large number of MTC (Machine - Type - Communication) devices in modern cellular mobile networks is a serious challenge as they can cause system overload for both RAN (Radio Access Network) and CN (Core Network). To address

this issue, various techniques have been developed that evaluate the network status (number of active devices) and thus better control RAN access.

Pseudo-Bayesian single-channel ALOHA (PB-ALOHA) [14] estimates the number of active devices based on the access result in the previous slot, but cannot adjust the transmission probability in a timely manner under heavily loaded traffic.

The Fast Adaptive S-ALOHA (FASA) method [15] has been developed for random access control of M2M systems with bursty traffic. Instead of observing in one slot, the statistics of consecutive slots of idleness and collisions are used in FASA to speed up the calculation of the number of active devices (network state). The FASA method is designed for single-channel systems. When calculating the number of active devices, it is assumed that the incoming flow is Poisson. However, this method works poorly under traffic loads with other distribution laws.

For multi-channel systems, the Multi-Channel Slotted ALOHA-Optimal Estimation (MCSA-OE) method [16] has been developed, which uses statistics from all channels to estimate the number of devices in each RA (Random Access) slot and does not depend on traffic models.

In [14], the number of active devices K is determined based on the Bayesian approach and Poisson traffic flow. If successful access is achieved, the value of K increases, otherwise it decreases.

The number of active devices can be determined based on the transmission of orthogonal preambles [17, 18]. It is assumed that all devices have unique preambles. A preamble is a sequence of bits of a certain length, which is much shorter than the length of a data packet. The base station must reliably determine the number of preambles in each channel. Active devices must transmit preambles before transmitting packets. The base station must send feedback information to active devices to report the number of transmitted preambles in several channels, which is used to estimate the number of active devices K. A formula for numerically calculating the channel capacity for an infinite number of preambles and a given finite number of channels has been obtained [17]. The probability of access p to a channel is defined as $p = 1/K$ [18]. The main disadvantages of these methods are the limited number of preambles and the possibility of collisions when transmitting preambles.

Reinforcement learning (RL) algorithms have become a popular method for random access and resource allocation in wireless networks [11]. Initially, Q-learning [19] and ALOHA-Q [20] algorithms were developed for single-channel random access. For multi-channel random access, deep Q-network (DQN) [21, 22] and deep reinforcement learning (DRL) [23–25] algorithms have been developed. Intelligent random access methods in sliced wireless networks (Sliced Mobile Networks) are proposed in [26]. DQN and DRL algorithms can adapt to changes in the environment and use past history to decentralize the learning of data transmission probabilities from devices. Intuitively, in a slowly changing environment, the longer the length of the state history, the better the decision an agent can make. However, a long history leads to an increase in the state space for the RL algorithm. When the state-action space is large and complex, deep neural networks are used to approximate the Q function. DRL algorithms are more efficient than RL algorithms. However, the complexity and power consumption are higher than those of RL algorithms. For most mMTC applications,

the main requirements are ease of implementation and low power consumption. The aim of this paper is to develop a simple RL-based multi-channel random access method with small states.

3 RL-based Multi-Channel Random Access Algorithm

One base station (BS) and N mMTC devices are used in the packet transmission process over the uplink of a single cellular network. Each slot in the network contains M resource blocks (RB), depending on the numerology (μ) that is utilized. Every gadget alternates between being inactive and active states. The device remains in the inactive state until it generates a packet according to Bernoulli's law with probability q. The active device with probability p accesses one of the M resource blocks to transmit data packets. The RB number is selected with equal probability. If a "success" occurs in the channel (RB) in which the device transmitted and the packet is transmitted, the subscriber switches from the active state to the inactive one. In case of a "conflict", the active device retransmits its packet in the RB of the next slot with probability p.

The following algorithms are to determine the value of the probability of the access.

1-persistent CSMA. The active device in each slot with probability 1 ($p = 1$) accesses RB and transmits a data packet. The algorithm consists of the following steps:

step 1. The activity of devices is determined. If $rand(1) \leq q$, then the device is active, otherwise it is passive.

step 2. The RB number is chosen at random by the active device using the probability $n_{rb} = randi(M)$.

step 3. The active device transmits a data packet to RB with the number r_{rb}.

step 4. If the data packet is transmitted successfully, i.e. there is no collision, then the active device goes into the passive state and step 1 is performed in the next slot.

step 5. If a collision occurs during the transmission of the data packet, then the active device remains in the active state and step 2 is performed in the next slot.

p-persistent CSMA. The active device accesses the RB and transmits a data packet with equal probability ($p \in [0, 1]$) in each slot. The algorithm consists of the following steps:

step 1. The activity of devices is determined: if $rand(1) \leq q$, then the device is active, otherwise - passive.

step 2. The access is determined by the active device using the probability $p = rand(1)$.

step 3. If $p \leq rand(1)$, then the active device selects the RB number $n_{rb} = randi(M)$, otherwise step 2 is performed in the next slot.

step 4. The active device transmits a data packet to RB with number n_{rb}.

step 5. If the data packet is transmitted successfully, i.e. there is no collision, then the active device goes into a passive state and step 1 is performed in the next slot.

step 6. If a collision occurs during the transmission of a data packet, the active device remains in the active state and step 2 is performed in the next slot.

RL-MCSA. To select the probability of access p to resource blocks, it is proposed to use the reinforcement learning method (RL), which is one of the mechanisms of artificial intelligence.

The problem of constructing systems that could adapt to the environment (in our case, the state of channels) and learn from the experience gained (collision and successful transmission statistics) has long attracted the attention of researchers from a wide variety of fields. The main components of RL and their interactions with the environment are shown in Fig. 1 [11].

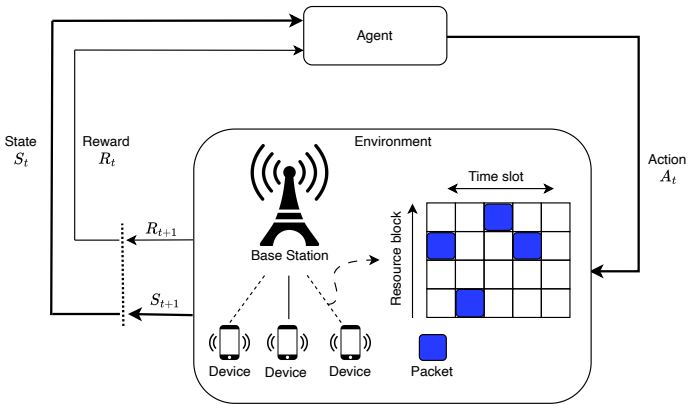


Fig. 1 The functional structure of the reinforcement learning-based MCSA algorithm.

An agent is an entity that makes decisions and performs actions in an environment. The environment is everything that surrounds the agent and with which it interacts. State (S) is the current position of the agent in the environment. Action (A) is the choice the agent makes in each state. Reward (R) is the feedback the agent receives after performing an action. The reward can be positive (upon achieving the goal) or negative. Q-function (Q-Value Function, Q) is a function that evaluates the quality of a state-action pair. It shows how profitable it is to perform a certain action in a given state. The Q-function is the basis for many reinforcement learning algorithms.

RL is a machine learning paradigm in which agents learn successful strategies that yield the greatest long-term benefit by interacting with the environment through trial and error. The most representative RL algorithm is the Q-learning algorithm [5]. Q-learning allows learning a good policy by updating an action-value function, called the Q-function, without an operational model of the environment.

The proposed access method is implemented as follows.

There are many states of channels (resource blocks) at time $(t-1)$

$$S(t-1) = \{s_0, s_1, s_2, ..., s_k, ..., s_M\} \tag{1}$$

where s_k is the total number of collisions in channels.

At time t the active device performs the action

$$a(t) \in A, \quad A = \{a_1, a_2, ..., a_K\} \tag{2}$$

where $a(t) - a_m$ corresponds to the event when the active device performs access with probability $p_m,\ m \in [1, K]$.

For each channel, the base station, upon successful transmission of a packet, sends an acknowledgment of correct reception (ACK) to the transmitting device, and upon detection of a collision, a request for re-transmission of the given data packet (NACK). Therefore, the feedback signal is

$$F(t) = \begin{cases} 1, & \text{on ACK;} \\ 0, & \text{on NACK.} \end{cases} \tag{3}$$

Each action at is rewarded (or penalized)

$$r(t) = u(t) - s(t) \tag{4}$$

where $u(t)$ and $s(t)$ are the total number of successful accesses and collisions at time t, respectively.

The goal of the system is to maximize the total reward. The Q-function is

$$Q(S(t-1), a(t)) = r(t) + \gamma \underset{a \in A}{max} Q(S(t+1), a) \tag{5}$$

where γ is the discount coefficient ($0 \leq \gamma \leq 1$).

In state $S(t-1)$, the active device at time t selects the action $a(t)$ (access probability) that has received the maximum reward up to time t.

In accordance with the above, the operation of **RL-MCSA** algorithm consists of the following steps:

step 1. The activity of devices is determined. If $rand(1) \leq 1$, then the device is active, otherwise – passive.

step 2. Active device i chooses access with the probability $p = \underset{a \in A}{max} Q(S(t-1, a(t)))$.

step 3. If $p \leq rand(1)$, then the active device selects the RB number $n_{rb} = randi(M)$, otherwise step 2 is performed in the next slot.

step 4. The active device transmits a data packet to RB with number n_{rb} and determines the reward $r(t)$.

step 5. If the data packet is transmitted successfully, i.e. there is no collision, then the active device goes into a passive state and step 1 is performed in the next slot.

step 6. If a collision occurs during packet transmission, the device remains in the active state and performs step 2 in the next slot.

We evaluate the efficiency of the algorithms by the following indicators. The probability of successful access is determined by the equation:

$$P_{sa} = N_{sa}/N_a, \tag{6}$$

where N_{sa} is the total number of successful accesses, N_a is the total number of accesses.

Radio channel utilization rate

$$K_{util} = N_{sa}/N_a, \tag{7}$$

where M is the total number of resource blocks.

The delay time for successful transmission of a data packet is determined by the following equation:

$$T = t_{sa} - t_{start}, \tag{8}$$

where t_{start} and t_{sa} are the moments of the start and successful transmission of a data packet, respectively.

The steps of the proposed algorithm are given in Fig. 2.

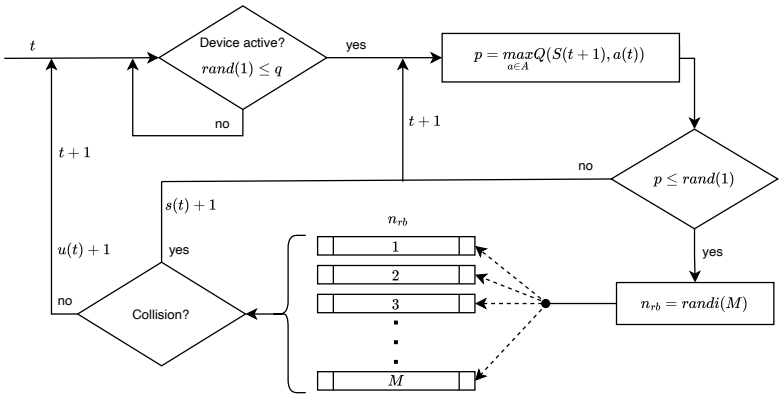


Fig. 2 The illustration of the RL-based MCSA algorithm.

4 Simulation Results

The simulation program is written in the MATLAB environment. The initial data for simulating the functioning of the access algorithms are: the total number of devices $N = 400$, the total number of resource blocks - $M = 30$, the probability of device activity $q = 0.3$, the discount coefficient $\gamma = 0$ and the slot duration $T_{slot} = 0.25$ mc.

Fig. 4 shows a graph of the change in the probability of successful access in each simulation cycle (K_{sc}) for $N = 120$.

The figure shows that the proposed RL-MCSA provides the highest probability of successful access compared to 1-persistent CSMA and p-persistent CSMA. The RL-MCSA does function as p-persistent CSMA up to cycle 100, collects statistics on the number of collisions and forms a reward matrix. After cycle 100, the device begins to select the access probability that has the maximum reward up to this cycle.

Figure 3 shows a graph of the dependence of the probability of successful access on the number of devices.

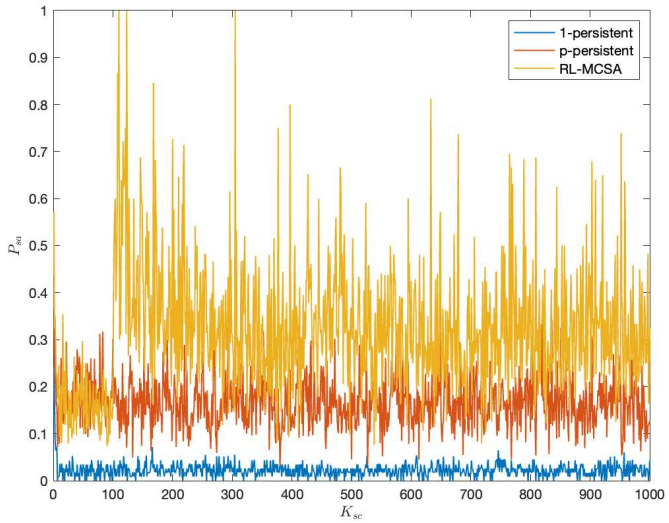


Fig. 3 Graph of the change in the probability of successful access in each simulation cycle for $N = 120$.

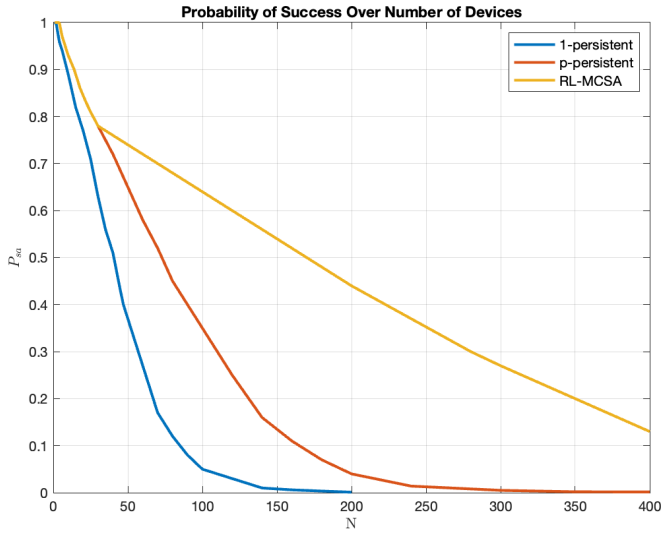


Fig. 4 Graph of the dependence of the probability of successful access on the number of devices.

As expected, the probabilities of successful access decrease with increasing number of devices due to the increase in the number of collisions. At the same time, the proposed RL-MCSA has an advantage in a wide range of changing the number of devices.

Fig. 5 shows a graph of the dependence of the radio channel utilization factor on the number of devices.

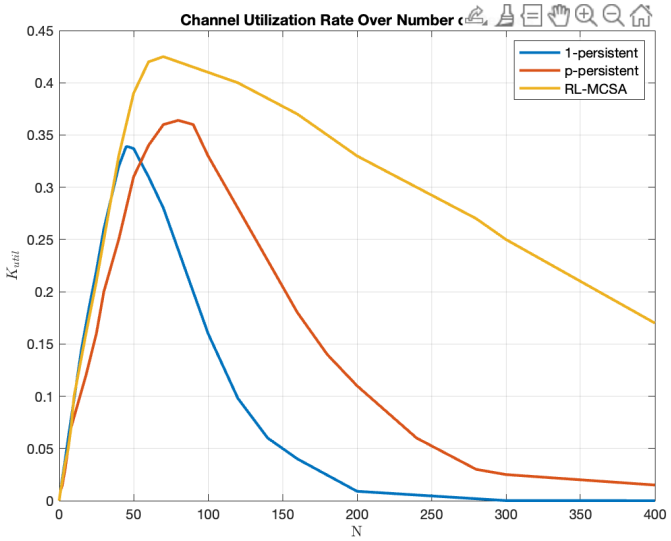


Fig. 5 Graph of the dependence of the radio channel utilization factor on the number of devices.

The utilization factor initially increases to the maximum value with the increase in the number of devices, and then decreases and tends to zero. This is explained by the fact that before the maximum point, the number of successful accesses is greater than the number of collisions, and after the maximum point, vice versa. The proposed RL-MCSA has the highest value of the maximum utilization factor. At the same time, after the maximum point, it decreases more slowly compared to 1-persistent CSMA and p-persistent CSMA.

Fig. 6 shows a graph of the dependence of the average delay time for successful transmission of a data packet on the number of devices.

It is obvious that the pattern of increasing average packet delay depending on the number of devices (Fig. 6) corresponds to the pattern of decreasing probability of successful access (Fig. 4).

Thus, the proposed reinforcement learning-based access algorithm improves the performance indicators of random access to the radio channel compared to 1-persistent CSMA and p-persistent CSMA.

5 Conclusions and Future Work

Random access methods are the main mechanisms for implementing the MAC layer of the 5G network in numerous mMTC applications. Various single-channel and multi-channel access methods are known in wireless networks, including effective random access methods in 5G networks based on deep learning mechanisms (DRL). Each

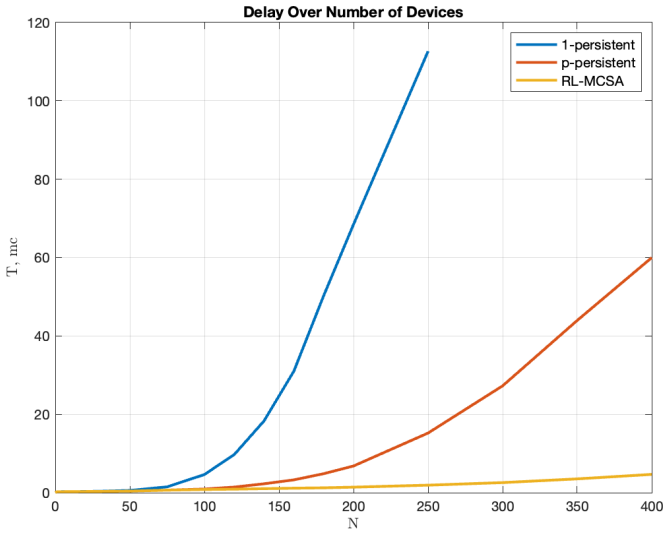


Fig. 6 Graph of the dependence of the average delay time for successful transmission of a data packet on the number of devices.

random access method has its own scope of application. In this paper, a multi-channel random access method based on reinforcement learning is proposed for mMTC applications, where the main requirement is the simplicity of implementing packet transmission devices. The simulation results show the effectiveness of the proposed method in terms of the probability of successful access and average packet delay time. In the next paper, the influence of self-similar traffic and the discount factor (γ) on the effectiveness of the random access method will be considered.

Declarations

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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